Nader Ayman

Malware Analyst | Reverse Engineer | Threat Intelligence Analyst

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Summary

Malware analyst and threat intelligence analyst who takes in-the-wild samples from the delivery chain to the final payload and turns them into intelligence: **config extractors, YARA, Sigma and Suricata detections, IOCs, MITRE ATT&CK mappings, infrastructure pivots and confidence-rated intelligence reports**. Published research on infostealers, RATs, ransomware and Android banking malware across x86/x64 PE, .NET, Go and Android, plus regional threat intelligence on phishing and ransomware targeting Egypt and the GCC. Experienced in static and dynamic analysis, memory forensics, threat hunting, passive infrastructure tracking and writing for both technical teams and executives.

Experience

Independent Malware & Threat Intelligence Researcher Apr 2026 – Present

Self-directed research, published at artfuldodger10.github.io and github.com/ArtfulDodger10 *Remote*

- Reverse engineered and documented 5 malware families from real campaigns (Lumma Stealer, XWorm, AgentTesla, KryBit ransomware, AntiDot Android banker) in an isolated lab with host-only detonation.
- Turned each case into finished intelligence: intelligence requirements, key judgments with confidence levels, Diamond Model, infrastructure pivots, and STIX 2.1 bundles managed in MISP and OpenCTI.
- Shipped reusable detection content: Python config extractors, YARA rules with 0 false positives on 145,000+ clean files, Sigma rules converted to Splunk SPL and KQL, Suricata rules, IOC sets and ATT&CK layers.
- Built an AI-assisted analysis workflow in which an LLM agent drives IDA Pro and x64dbg under guard rules, with evidence-tagged findings and a fact-check pass before anything is published.

Cyber Security Instructor Jul 2024 – Aug 2025

Udacity / EYouth — Part-time *On-site*

- Mentored 100+ students in cybersecurity fundamentals, malware analysis and threat detection; reviewed projects and gave technical feedback.
- Designed and led workshops on reverse engineering methodology, Sysmon telemetry and threat hunting built around real-world detection scenarios.

Threat Intelligence Projects

Lumma Stealer Campaign Intelligence: From Config to Infrastructure *In progress* Oct 2026

- *RDAP, crt.sh, passive DNS, urlscan, Shodan, Censys, ThreatFox, MalwareBazaar, MISP, OpenCTI, STIX 2.1*
- Enriched the 17 C2 domains recovered by my extractor from passive sources only, and clustered them by registrar, name servers, TLS certificates, hosting ASN and registration timing.
- Wrote urlscan and Censys hunting queries from the shared traits and measured each query's precision (hits vs. confirmed C2s); pivoted on campaign IDs across ThreatFox and MalwareBazaar to build a timeline.
- Produced a Diamond Model per delivery chain, key judgments with confidence levels and stated gaps, and a STIX 2.1 bundle imported into MISP and OpenCTI.

Phishing Infrastructure Targeting Egyptian Banks and Fintechs *In progress* Nov 2026

- *crt.sh, CertStream, urlscan, dnstwist, RDAP, Shodan, Censys, Python, STIX 2.1*
- Tracked lookalike domains impersonating five Egyptian financial brands over a fixed collection window using passive sources only; clustered them by certificates, hosting, registrars and phishing-kit traits.
- Released an open-source Python lookalike-domain monitor (certificate transparency, dnstwist, urlscan) with STIX 2.1 output; reported live phishing sites to the brands and hosting providers before publication.

Ransomware Threat Brief: Egypt and GCC *In progress* Nov 2026

- *ransomware.live, MITRE ATT&CK, IDA Pro, YARA, Sigma*
- One-page BLUF brief for CISOs on a ransomware group active against Egypt and GCC organizations: victimology from leak-site data (treated as claims, not confirmed breaches), TTPs from my own reversing of its encryptor, detections and prioritized recommendations.

Malware Analysis Projects

Lumma Stealer: Two Delivery Chains, One Config Extractor | *GitHub* Oct 2026

- *IDA Pro, x64dbg, Volatility 3, Python, YARA, FakeNet-NG, Process Hacker*
- Reversed two unrelated loaders end to end: a Go RunPE loader that hollows Lumma into `explorer.exe`, and a trojanized "AIO Checker" tool chaining an RC4 C++ crypter, a dropper and a .NET crypto clipper before hollowing Lumma into `MSBuild.exe`.
- Recovered the payload's protections statically: ChaCha20-encrypted C2 table, MBA-obfuscated stack strings, FNV-1a hashing, Steam dead-drop resolver, Russian-locale geofence and hypervisor check.
- Built static unpackers and one config extractor for both chains, from disk or from a memory dump of the hollowed process, validated on all 7 campaign samples; released YARA rules, IOCs, an ATT&CK layer and two reports.
- *In progress:* memory forensics of the hollowed processes with Volatility 3 (malfind, yarascan, handles), confirming the injected payload and the clipper's mutex in memory.

XWorm V7.1: From Phishing Email to RAT | *GitHub* Jul – Sep 2026

- *dnSpy, x64dbg, PowerShell, Python, YARA, Suricata, Wireshark*
- Traced a phishing campaign from a RAR attachment through a JavaScript dropper (AES-256-CBC, ChaCha20, XOR, gated behind fake-activity counters) and a PowerShell loader with a custom base52 decoder to XWorm loaded reflectively into a hollowed `MSBuild.exe`.
- Broke the config encryption (AES-256-ECB keyed by an offset-duplicated MD5 of the mutex) and wrote a static config extractor tested on 9 builds from V3.0 to V7.4, plus a MalwareBazaar hunting script.
- Wrote Suricata rules for the XWorm TCP C2 protocol and tested them against the captured traffic.

KryBit Ransomware (Babuk-derived): Encryptor Reversing *In progress* Oct 2026

- *IDA Pro, capa, FLOSS, YARA, Sigma*
- Confirmed the family from its file extension and mutex rather than AV labels; identified the hybrid encryption scheme (Curve25519 key exchange, HC-128 file encryption, SHA-512 key derivation, CRC-32 key check) and the encrypted-file footer layout.
- Mapped recovery inhibition (shadow copy deletion, service and process termination, hidden volume mounting), command-line options, network share enumeration and exclusion lists; compared the build against published Babuk analyses; wrote and tested 2 YARA and 2 Sigma rules.

AntiDot Android Banking Trojan: Threat to Egyptian Banking Customers *In progress* Oct 2026

- *jadx, apktool, Frida, IDA Pro (ARM64), Python, MITRE ATT&CK for Mobile*
- Reversed a native-backed Android packer: XOR-decoded asset manifest, per-ABI native library loaded with `System.load`, hidden DEX decrypted in native code and dumped with Frida; reimplemented the string decryption in Python.
- Documented the banker's abuse of Accessibility services, overlays, the default SMS role (OTP theft), call screening and screen capture; extracted C2 and the target-app list, mapped to ATT&CK for Mobile with detections and recommendations for banks.

AgentTesla: Four-Stage .NET Loader to SMTP Credential Stealer | *Report* Jun 2026

- *dnSpy, x64dbg, FakeNet-NG, Process Monitor, Wireshark, YARA, Splunk*
- Unpacked four .NET stages (XOR-encoded resource, ConfuserEx loader with a payload hidden in image pixels, DeepSea 4.1 process hollowing) and recovered the SMTP exfiltration config; released a config extractor, YARA rules, a Splunk hunting query, IOCs and an ATT&CK layer.

Process Injection Telemetry Research: WinAPI vs. Direct vs. Indirect Syscalls | *GitHub* May 2026

- *C, x64 Assembly, Sysmon, Sigma, Splunk SPL, KQL, PowerShell*
- Implemented three T1055 process injection variants from scratch with the same payload and target, so each telemetry difference comes from the injection method alone; wrote three Sigma rules from live Sysmon telemetry (GrantedAccess masks, CallTrace, StartModule) and converted them to Splunk SPL and Microsoft Sentinel KQL.
- Documented five findings (e.g. unexpected Event ID 8 for indirect syscalls), corrected two widely cited documentation errors, and published false-positive tuning and a detection gap analysis (stack spoofing, BYOVD).

YARA Detection Rules | *GitHub* 2026 – Present

- Family and campaign rules for XWorm, AgentTesla, AsyncRAT, Lumma and KryBit, for files and memory dumps, with the reason for every string; 0 false positives across 145,000+ clean Windows files, true-positive tests per family, and GitHub Actions CI that compiles and tests every rule on each commit.

Manual Unpacking and CTF Reverse Engineering Writeups | *Blog* | *Medium* 2025 – 2026

- Manual UPX and custom packer unpacking (OEP, memory reconstruction, IAT repair) and reverse engineering writeups; competed nationally with team Sp3c0ps, ranked Top 5 in Egypt.

Technical Skills

Threat Intelligence: intelligence lifecycle and requirements, MITRE ATT&CK (Enterprise and Mobile), Diamond Model, Cyber Kill Chain, Pyramid of Pain, MISP, OpenCTI, STIX 2.1, TAXII, IOC and TTP enrichment, infrastructure pivoting, threat actor profiling, OSINT, dark web and Telegram monitoring, brand protection, confidence-rated reporting, executive briefings

Intelligence Sources: VirusTotal (hunting), MalwareBazaar, ThreatFox, URLhaus, urlscan, Shodan, Censys, crt.sh, CertStream, RDAP/WHOIS, passive DNS, ransomware.live, dnstwist

Reverse Engineering: IDA Pro, IDAPython, Ghidra, Binary Ninja, x64dbg, WinDbg, dnSpy, ILSpy, de4dot, jadx, apktool, Frida, Radare2, PE-bear, Detect It Easy, capa, FLOSS, pefile

Malware Analysis: static and dynamic analysis, unpacking, packers and crypters, process injection and hollowing, API hashing, config and string decryption, anti-analysis, crypto identification (AES, RC4, ChaCha20, HC-128, Curve25519), PE, .NET, ELF, DEX and APK formats

Sandboxes and Forensics: ANY.RUN, Hatching Triage, CAPE, Joe Sandbox, REMnux, FLARE-VM, FakeNet-NG, INetSim, Sysinternals (Process Monitor, Process Explorer, Autoruns), Volatility 3, Wireshark, Zeek

Detection Engineering and Threat Hunting: YARA, Sigma, Suricata, Snort, Sysmon, ETW, Splunk (SPL), KQL (Microsoft Sentinel, Defender, Elastic), SIEM, EDR telemetry, detection-as-code, false-positive tuning

Programming: Python, C, C++, C#, x86/x64 Assembly, PowerShell, Bash; reading Go, ARM64, JavaScript, VBScript

Systems: Windows Internals, Linux, VMware, VirtualBox, Docker, Git, GitHub Actions

Certifications and Training

Certifications: Cyber Threat Intelligence 101 (CTI101), arcX

Training: Advanced Malware Analysis (Mahara-Tech); Reverse Code Engineering (Mahara-Tech); Cyber Security Training Camp (Information Technology Institute, ITI); eJPT preparation (Netriders Academy); RHCSA preparation (Mahara-Tech); TryHackMe Cyber Threat Intelligence path

OpenSecurityTraining2: Architecture 1001 (x86-64), Symbolic Analysis, Reversing C++ Binaries, GDB

Self-study: Practical Malware Analysis; Mastering Malware Analysis; Windows Internals; Visual Threat Intelligence (Roccia); Intelligence-Driven Incident Response (Roberts & Brown)

Education

Minufiya University Minufiya, Egypt
B.Sc. Computer Science and Engineering *Oct 2021 – Feb 2026*

- **Relevant Coursework:** Operating Systems, Network Security, Cryptography, Cyber Security, Computer Architecture, Cloud Computing

Languages

Arabic: Native **English:** Professional Working Proficiency